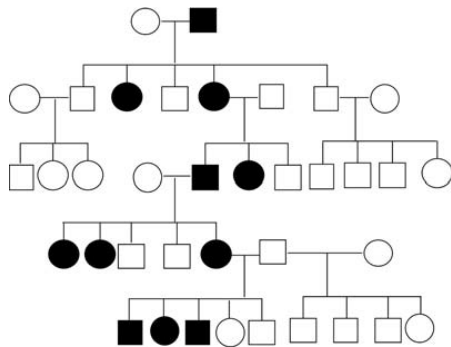


- How are continuous traits different from discrete traits? Explain. (2 marks)
- In mice, two independently assorting genes control coat colour and tail length. The Q allele gives yellow coat colour and is dominant to grey coat colour (q) allele. The R allele gives short tail and is dominant to the normal tail length (r) allele. Both (Q) and (R) are recessive lethals, killing the zygote. If two mice, both with Yellow coat and short tail are crossed, what proportion of the offspring is expected to have a grey coat and a normal tail? (2 marks)
- The broad sense heritability of bristle (hairs) in fruit flies is 0.5. A certain population of fruit flies shows a mean bristle number of 126 with a standard deviation of 6. Calculate the components of variation (2 marks)
- Assume that in a dihybrid cross starting from pure breeding Black and Golden parents, the F₂ shows a 9:3:4 ratio for Black: Brown: Golden phenotypes. If we take the F₁ individual from the above cross and subject it to a test cross, what is the expected phenotypic ratio? Explain. (3 marks)
- "In X linked loci, the alleles are neither dominant nor recessive. They are just X linked". What is your opinion about this statement? Explain (2 marks)
- Consider the following pedigree.



- What is the most probable mode of inheritance? (Autosomal or X linked/ Dominant or recessive). With reference to the pedigree, point out and explain the strongest evidence in favour of your answer. (1+ 2 marks)
- If the individual V-3 marries a normal person, what is the probability that their first child will be a son who express the trait? (2 marks)

- Trihybrid geranium plants with hairy leaves, blue flowers, and tall-growth habit are testcrossed with plants having smooth leaves, red flowers, and short-growth habit. The progeny are listed in the table.

Leaf	Flower	Height	Number
Hairy	Blue	Tall	368
Hairy	Blue	Short	20
Hairy	Red	Tall	1,797
Hairy	Red	Short	179
Smooth	Blue	Tall	191
Smooth	Blue	Short	1,752
Smooth	Red	Tall	26
Smooth	Red	Short	340
			<u>4,673</u>

- Which are the Parental, Single, and double cross overs? (2 marks)
- What is the gene order? (2 marks)
- Create a linkage map of these three loci. (3 marks)

BIO 102: Mid Semester Examination 1
Total Marks: 23

Registration Number:
Name: